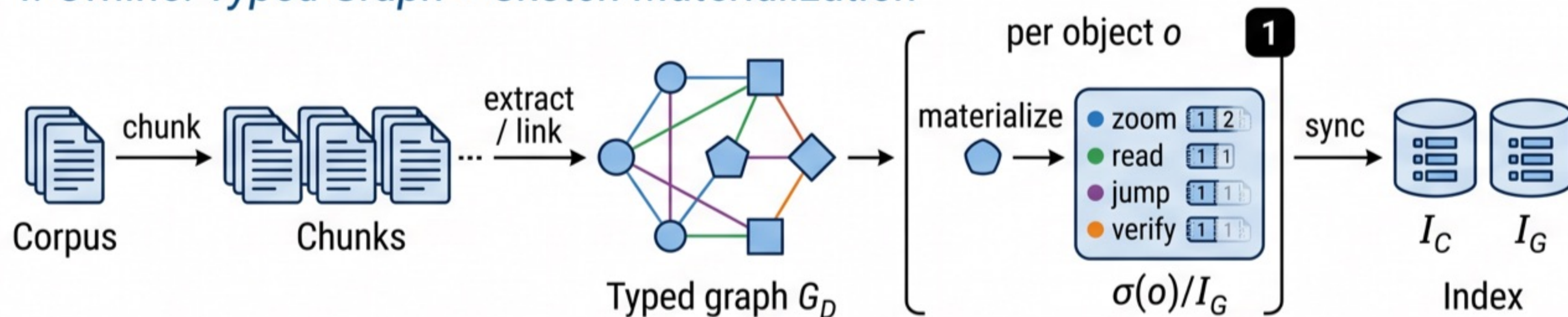
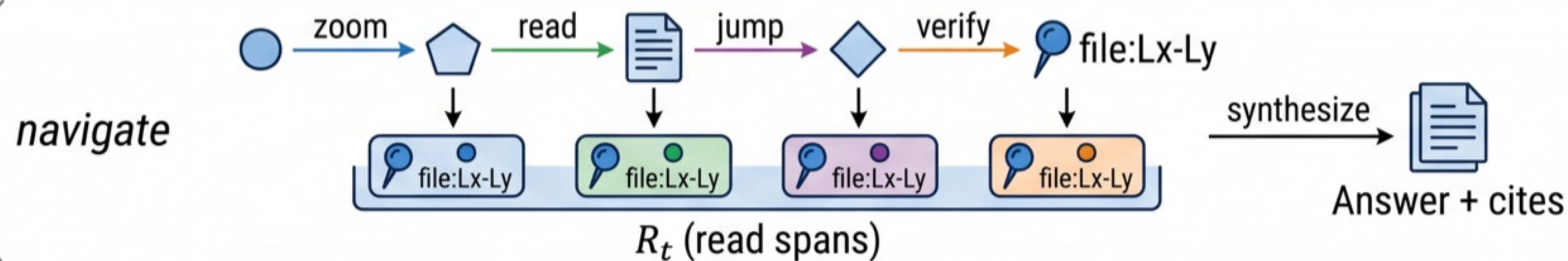
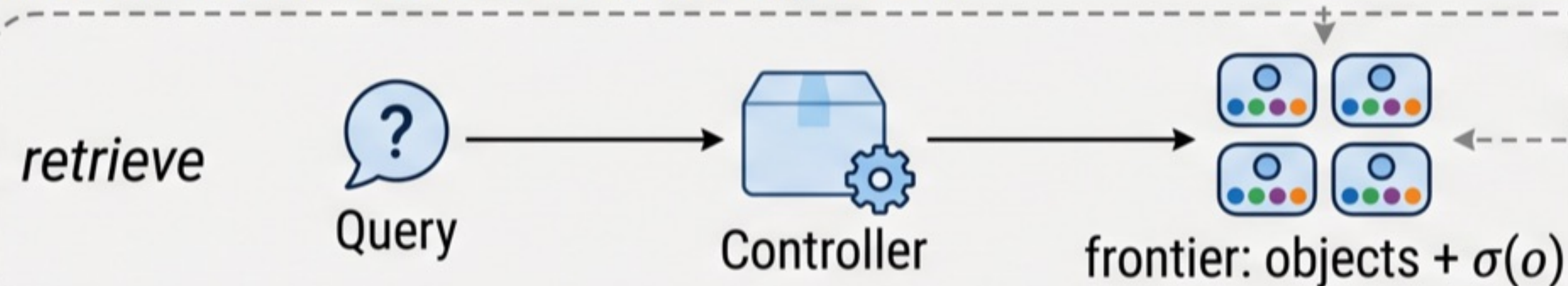


I. Offline: Typed Graph + Sketch Materialization

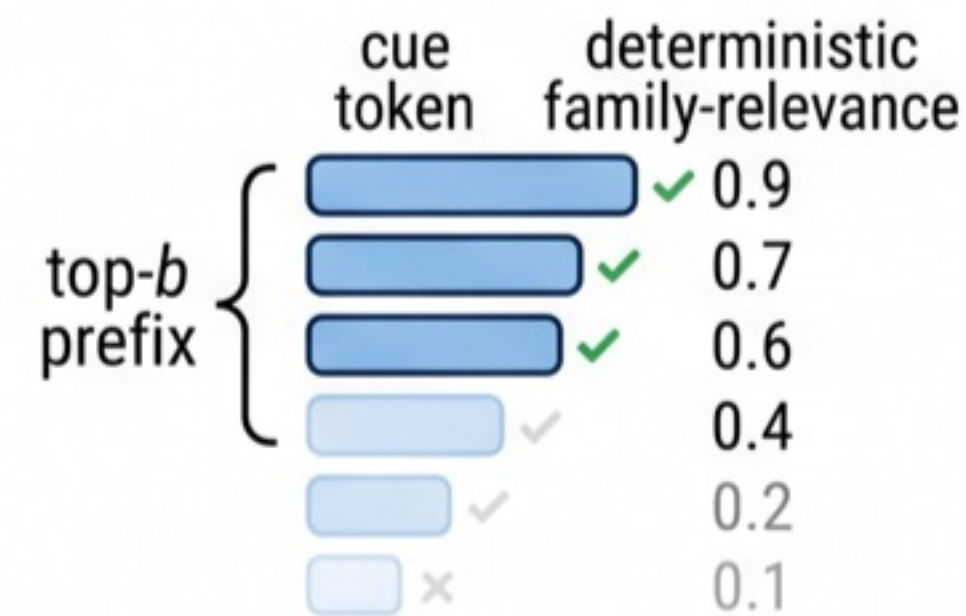


II. Online: Sketch-Lookup Navigation

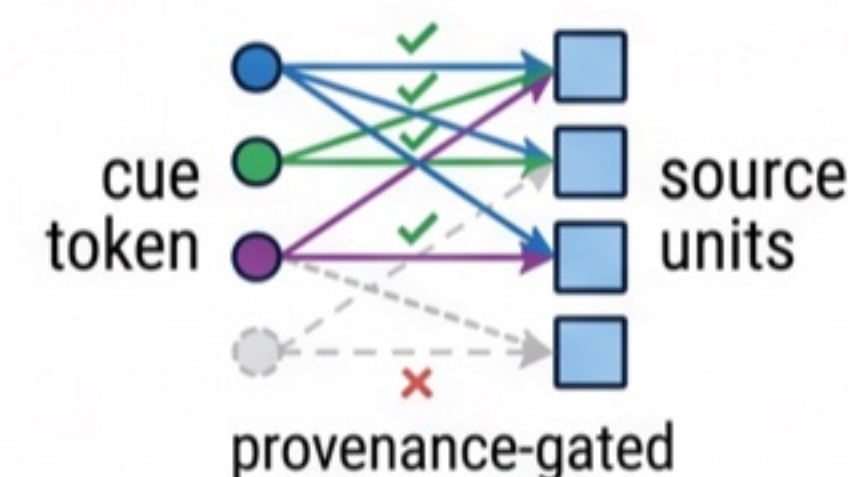


follow typed cues by object lookup + source read (no graph re-expansion)

Cue materialization (deterministic ranking) 1



default: complete (no cap)



Bounded per-step exposure 2

